

The Perturbation Catalogue:

An Integrated Resource for Exploring the Impact of Genetic Perturbations

Kirill Tsukanov • Aleksandr Zakirov • Alexey Sokolov • Mallory Freeberg
EMBL European Bioinformatics Institute, Wellcome Genome Campus, Hinxton, United Kingdom

Large-scale perturbation studies reveal gene function, disease mechanisms and candidate drug targets, but their evidence remains fragmented across repositories, formats, identifiers and study-specific processing choices. The Perturbation Catalogue brings these experiments into a shared, searchable resource while retaining the biological and technical context needed to interpret every result.

Three complementary scales of evidence

For a target of interest, the Catalogue links **gene-level screening phenotypes**, variant-level functional measurements and single-cell transcriptomic responses.

What each modality contributes

	Perturb-Seq	CRISPR screens	MAVE
Resolution	Single cell; gene perturbation × transcriptome	Gene perturbation × phenotype or fitness	Amino-acid or nucleotide variant × functional score
Primary readout	Cell-by-gene expression counts	Gene-level depletion, enrichment or assay score	Variant effect measurements across protein sequence
Catalogue analysis	Standardised differential expression and Hallmark gene-set enrichment	Harmonised author-generated scores and significant hits	Harmonised author-generated scores and variant-effect heatmaps
Explore by	Perturbed target, affected gene, dataset and biological context	Perturbed target, dataset and biological context	Target, variant position, amino-acid change and dataset
Current production coverage	26 datasets ; five reprocessed from raw FASTQ	1,201 datasets	10 datasets

Why integration is difficult

A published result is inseparable from its experimental design. The same perturbation can produce different effects across model systems, assay endpoints, delivery methods, time points and analysis pipelines. Gene symbols change; repositories expose different identifiers; metadata depth varies; and useful context is often available only in the paper or supplement. The Catalogue therefore preserves source provenance while converting data and metadata into common structures. The goal is not to erase genuine biological differences, but to remove avoidable friction before comparison and reuse.

Metadata that carries meaning

Each curated dataset is validated against a unified, extensible schema. Core domains include:

Study

Title, authors, accession, publication year, source repository and licence.

Perturbation

Mechanism, library generation, enzyme and library delivery, treatment and time point.

Assay

Readout, sequencing platform, software, score interpretation and processing provenance.

Model system

Tissue, cell type or cell line, disease, sex and developmental stage.

Target

Ensembl gene ID, approved symbol and name, biotype and genomic identity.

Biological entities are aligned to established ontologies wherever possible. Curation also reveals missing or inadequate terms, allowing the project to contribute improvements back to ontology communities.

Production catalogue at a glance

1,237 datasets

36 tissues

1,200 cell lines

Production snapshot: 24 August 2026.

19,235 targets

34 cell types

169 diseases

How evidence enters the Catalogue

Public repositories and published studies are curated into modality-specific inputs, validated, normalised and transformed into shared analytical and metadata tables. The same backend and API then serve every modality.

sources → curation + validation → BigQuery + dbt → Elastic + Postgres → API / portal / files

shared schemaharmonised tablessearch + data

Ensembl gene IDs provide the stable target identity across ingestion, search and retrieval. Dataset-level provenance, licence and score interpretation remain visible rather than being detached from the measurements they qualify.

Perturb-Seq reprocessing from raw FASTQ

Author-provided expression matrices differ in alignment software, gene annotation, filtering and guide assignment. Our reprocessing workflow starts again from raw sequencing reads so that selected studies share one computational path:

FASTQ → cell × gene counts → guide assignment → QC → DEA → GSEA

Differential expression records the direction and magnitude of gene-level response to each perturbation. Gene-set enrichment summarises coordinated changes using MSigDB Hallmark pathways. Final data, summaries and provenance are published through the same Catalogue interface and API.

Five reprocessed datasets are live:

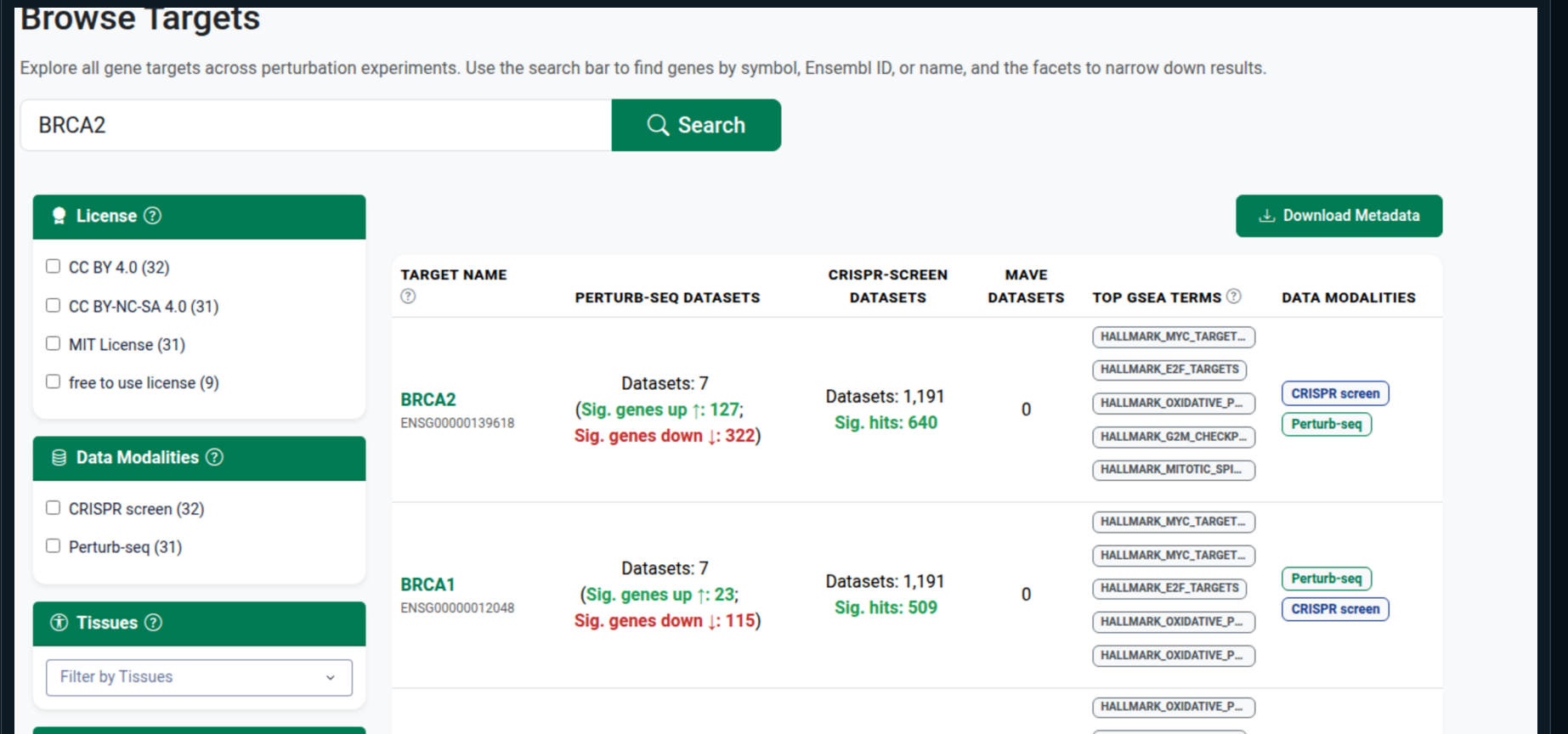
- Nadig 2025: Jurkat and HepG2
- Replogle 2022: K562 essential, K562 genome-wide and RPE1 essential

What “comparable” means here

Shared processing reduces avoidable technical variation; it does not make distinct experiments biologically equivalent. Current validation compares broad agreement with author-generated matrices where available. Developing author-independent quality metrics and wider kit support remains an active part of the work.

Explore targets and datasets

Search a gene once to see which modalities contain evidence, which datasets contribute it, significant perturbation effects and top enriched pathways. Dataset pages expose the full curated context and distinguish author-provided from reprocessed data.



Live target browser for BRCA2: modality coverage, significant hits and Hallmark enrichment terms.

Available now

- Search and filter targets or datasets across all three modalities.
- Inspect curated experimental context, ontology links, licence and provenance.
- Explore Perturb-Seq differential expression and pathway enrichment.
- View significant CRISPR hits and MAVE variant-effect heatmaps.
- Use Ensembl gene IDs consistently across modalities.

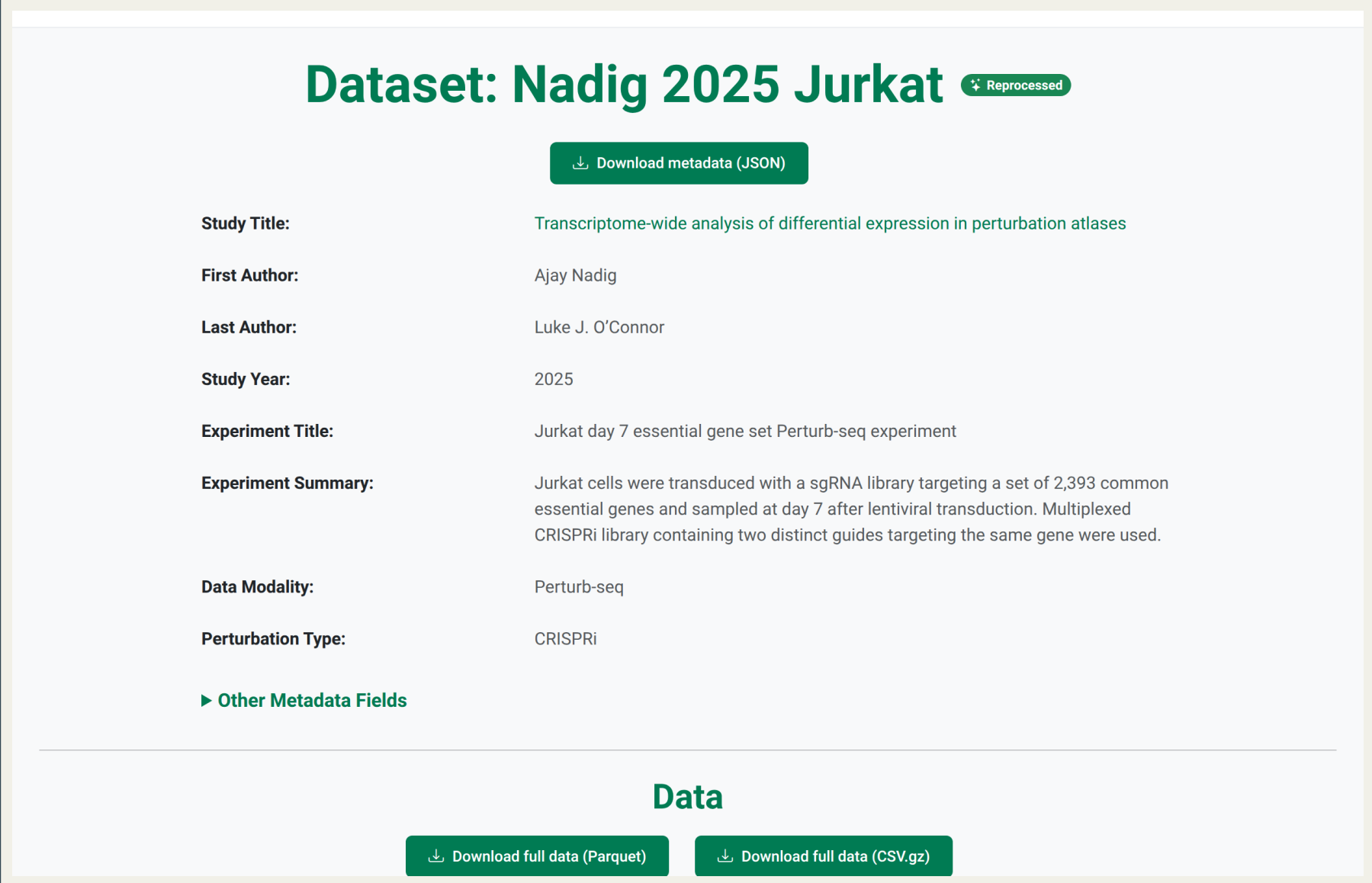
API and complete downloads

Everything displayed by the portal is available programmatically. Common entry points include:

```
GET /search?query=BRCA2&search_mode=targets
GET /dataset/{dataset_id}
GET /v1/perturb-seq/{dataset_id}/search
```

Full datasets are precomputed as **Parquet** and gzip-compressed **CSV**; dataset metadata is available as **JSON**. Filtered table views can also be downloaded directly, making the Catalogue useful both interactively and inside reproducible workflows.

Inside a reprocessed Perturb-Seq dataset



Interpretation begins before the first result

The dataset page identifies the source study, experiment, modality and perturbation mechanism, and labels data that have been reprocessed from raw sequencing reads. Additional metadata remains one click away rather than being reduced to a detached accession number.

From summary to the complete evidence

Interactive rows connect perturbations and effect genes to Ensembl IDs, effect size, adjusted significance, statistical score and biological context. The same page provides complete Parquet and compressed CSV files—not only the rows visible after filtering.

Metadata is data, too

The full validated metadata record is downloadable as JSON. This makes experimental context and provenance available to computational workflows, supports reproducible reuse and exposes exactly how a dataset was interpreted by the Catalogue.

Production example: Nadig 2025 Jurkat. The reanalysis targets 2,393 common essential genes in a day-7 CRISPRi experiment.

What we are building next

Curator-reviewed metadata extraction; broader, modality-appropriate reanalysis; support for additional datasets and experimental designs; and periodic, versioned releases with explicit provenance.

Work with us

Suggest a dataset, challenge our schema or request a feature. Funded multi-year collaborations can prioritise the modalities, datasets, analyses or functionality most relevant to your programme.
perturbation-catalogue-help@ebi.ac.uk

Explore the Catalogue



EMBL-EBI

Open Targets